Vision Document

Course Instructor

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1) Document Version

Version	Date	Description	Authors
V1.0	February 25, 2026	Initial Vision Document	Burhan Ahmed Ali Akbar Huzaifa Ali
V2.0	February 27, 2026	Refined Vision Document	Burhan Ahmed Ali Akbar Huzaifa Ali

2) Introduction

2.1 Purpose

The Vision Document serves as a tool to establish and record the essential requirements and functions of SmartKiryana which operates as a digital management solution for Pakistani grocery stores that have a small to medium size operation.

The document provides information about the system stakeholders and essential system capabilities and business goals of the system. The document describes the difficulties traditional kiryana stores encounter while explaining the solutions provided by SmartKiryana to solve those problems. The Use Case and Supplementary Specifications documents will provide a comprehensive description of system operations and its operating procedures.

2.2 Scope

SmartKiryana operates as an all-inclusive Grocery Store Management System which enables digital transformation for small and medium-sized supermarkets and kiryana stores throughout Pakistan. The present situation shows that numerous stores depend on traditional paper-based systems to document:

- Inventory
- Sales
- Orders
- Employee performance

These methods are inefficient consequently store owners face:

- Overstocking and understocking issues
- Incorrect billing
- Audit failures
- Poor employee performance tracking
- Limited business analysis capabilities

SmartKiryana will provide:

- Role-based dashboards for Admin and Employees
- Automated inventory management
- Sales analytics and reporting
- Employee performance tracking
- Target-based bonus management
- Billing through barcode, serial number, or manual search

The system aims to improve operational efficiency, reduce financial losses, and enable data-driven decision-making for grocery store owners.

2.3 Definitions, Acronyms, and Abbreviations

- **POS:** Point of Sale
- **Admin:** Shop Owner or Store Manager
- **Employee:** Salesperson or cashier
- **SmartKiryana:** The proposed grocery store management system

3) Business Opportunity

The retail grocery sector in Pakistan consists largely of small and medium sized kiryana stores that rely on manual bookkeeping or outdated POS systems. The stores face operational challenges because they compete in markets where their success depends on maintaining efficient operations and controlling costs and managing their inventory systems.

The country needs a management system that can operate at local price points while providing simple functionality to meet the operational needs of grocery stores throughout Pakistan.

The system aims to:

- Reduce inventory losses caused by overstock and stockouts
- Improve billing accuracy
- Provide business insights through sales analytics
- Enhance employee accountability through performance tracking
- Enable shop owners to make data-driven decisions

By digitizing store operations, SmartKiryana creates measurable financial and operational benefits for small business owners.

4) Problem Statement

Small-to-medium grocery shops in Pakistan are inefficiently operated since they maintain all records manually and are lacking in automated systems.

These issues are:

- Inaccurate billing
- Inventory mismanagement
- Overstocking and understocking
- Lack of sales analysis
- Poor employee performance tracking
- Limited audit transparency

As a result, shop owners experience financial losses, poor decision-making, and reduced operational control.

SmartKiryana solved the existing problems through its complete management solution which automatically handles billing processes and inventory management, performance monitoring and sales data collection.

5) Summary of Features

5.1 Database Management

SmartKiryana will use a centralized relational database to make operations secure and efficient.

The database shall:

- Store product details (barcode, name, category, price, supplier, stock quantity)
- Maintain transaction history
- Record employee credentials and roles
- Store sales targets and bonus records
- Maintain return records
- Support audit tracking

The system will ensure data integrity, backup support, and secure access control.

5.2 Account and Access Management

The system will implement role-based access control for the separation of administrative and operational responsibilities.

5.2.1 Admin Account

The admin of the system will be able to:

- Create and manage employee accounts
- Assign login credentials
- Set monthly sales targets
- Define bonus structures
- Manage Inventory
- Access full sales analytics
- Monitor stock alerts
- Review employee performance

5.2.2 Employee Account

All employees of the shop will be able to:

- Log into personal dashboards
- Process sales transactions

- Edit bills before checkout
- Handle item returns
- View assigned targets
- Track personal sales history

5.3 Billing and Sales Management

This handles customer transactions.

Features include:

- Add items via barcode scanning
- Add items via serial number
- Manual item search
- Modify quantities before checkout
- Generate printed or digital receipts
- Maintain complete bill history

The system will automatically update inventory upon transaction completion.

5.4 Inventory Management

SmartKiryana provides automated stock control for inventory management.

Features include:

- Automatic stock deduction after a sale has completed
- Automatic stock restoration after a return has been completed
- Understock alerts
- Overstock alerts
- Real-time stock monitoring
- Item categorization

This ensures that inventory levels are accurate and prevents over/under stocking.

5.5 Sales Analytics and Reporting

The system provides built in analytics to allow data driven decision making.

Features include:

- Filter sales by month

- Filter sales by revenue
- Identify top-selling products
- View frequency of sales per item
- Revenue trends analysis
- Exportable reports

5.6 Employee Performance and Target Management

SmartKiryana provides performance tracking of employees to monitor sales and efficiency and increase motivation.

Features include:

- Monthly sales target setting
- Automated bonus calculation
- Employee sales comparison
- Individual performance reports
- Historical performance tracking

6) Product Overview

SmartKiryana serves as a digital Grocery Store Management System which specifically meets the operational needs of small and medium-sized kiryana stores throughout Pakistan. The system replaces manual bookkeeping and outdated POS systems through its complete solution which handles billing and inventory management and employee performance tracking and sales analytics on one platform.

SmartKiryana will be:

- A desktop-based client-server application
- Designed with a simple and user-friendly interface
- Optimized for low-resource environments
- Capable of operating in offline mode with data synchronization

The product aims to improve efficiency, reduce financial losses, enhance accountability, and enable data-driven decision-making for shop owners.

7) Stakeholders

7.1 User Stakeholders

7.1.1 Admin

The admin is the primary decision-maker and system controller.

The admin is responsible for:

- Managing employee accounts
- Managing Inventory
- Monitoring sales and revenue
- Setting monthly sales targets
- Reviewing employee performance
- Monitoring inventory alerts
- Making business decisions based on analytics

The admin gains improved control, transparency, and business insights.

7.1.2 Employee

Employees of the store use system to process customer sales and keep check on their own performance.

Their responsibilities are:

- Processing customer sales
- Handling returns
- Viewing their sales targets
- Track their own performance

Employees benefit from simpler billing and performance tracking.

7.2 Non-User Stakeholders

7.2.1 Development Team

Responsible for creating and maintaining the system.

7.2.2 Government Regulators

Regulation officers would like to view revenue for taxation purposes.

7.2.3 Store Customers

Indirect stakeholders of the system who receive faster, accurate billing and improved service.

7.2.4 System Administrator / IT Support

Responsible for services such as maintenance, updates, backups, and troubleshooting of the system.

8) Dependencies and Constraints

8.1 Dependencies

SmartKiryana depends on:

- Availability of compatible hardware (computer, barcode scanner, printer, database)
- Reliable power supply
- Internet connection (for updates and synchronization)
- Basic computer literacy of employees

8.2 Constraints

Constraints are:

- The system must be available 24/7.
- The system must have high usability.
- The system must be highly accurate having 99.9% accuracy.
- The system must be highly secure.

9) Cost and Pricing

9.1 Cost

Item	No. of items	Cost (PKR)	Total (PKR)
Developers	3	80,000	240,000
Printer	1	25,000	25,000
Scanner	1	15,000	15,000
Other	-	20,000	20,000

9.2 Pricing

Pricing Strategy:

SmartKiryana will either be a:

- One-time installation package for small stores **OR**
- Subscription-based model (monthly fee)

The system will remain much more affordable compared to other POS systems while remaining competitive and up to par.

10) Licensing and Installation

SmartKiryana will be licensed for commercial use under a proprietary license model.

Installation includes:

- Software deployed on store computer system
- Database configuration
- Hardware to be used by system such as printer, barcode scanner
- Employees must be trained to use system
- User manual to aid administration.

The system will support:

- On-site installation
- Remote setup assistance
- Periodic updates and maintenance